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1.3 Problem 1(c) *(5 Points)* 9

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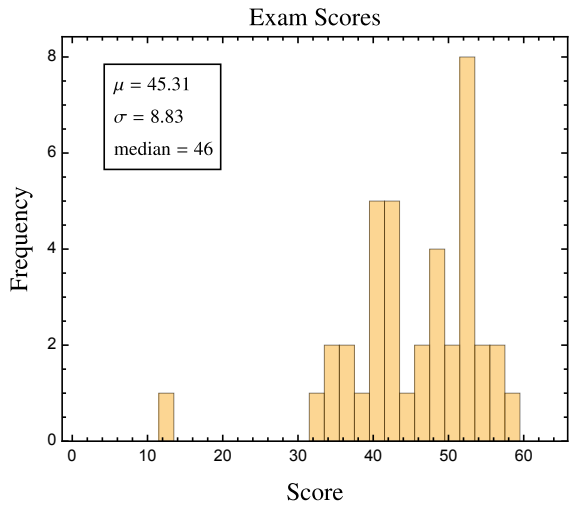
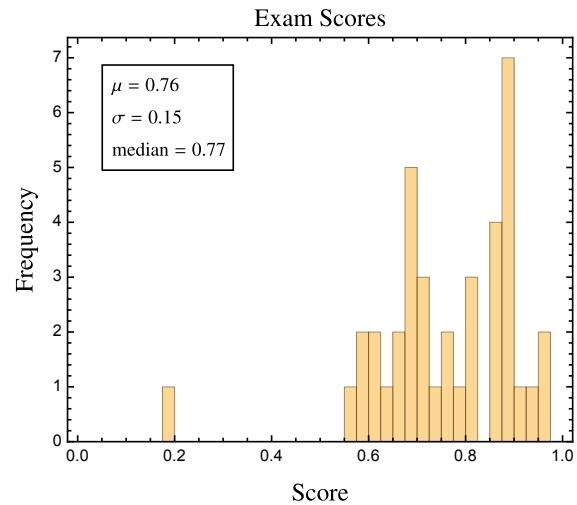


(a) The distribution of scores on Midterm 2



(b) The distribution of % scores on Midterm 2.

Figure 1: Midterm 2 scores

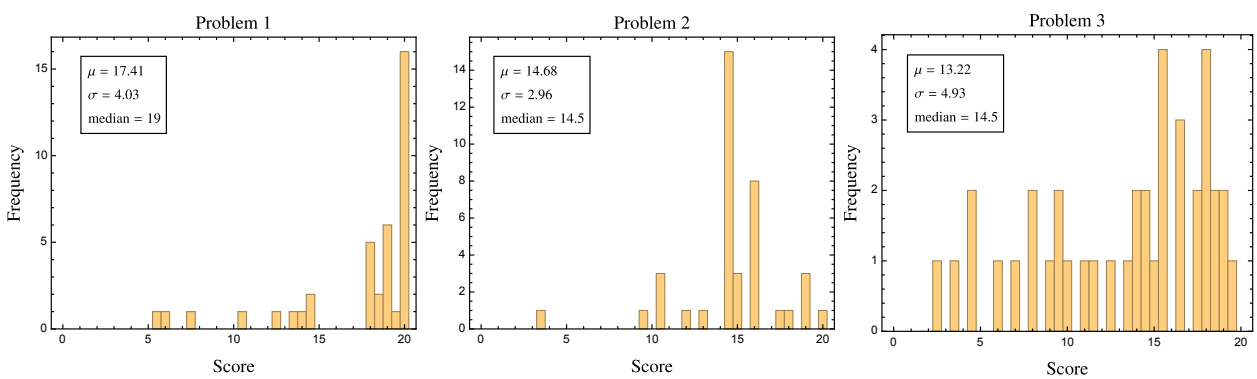


Figure 2: Breakdown of distributions by problem

0.1 Summary Histogram: Problem 1

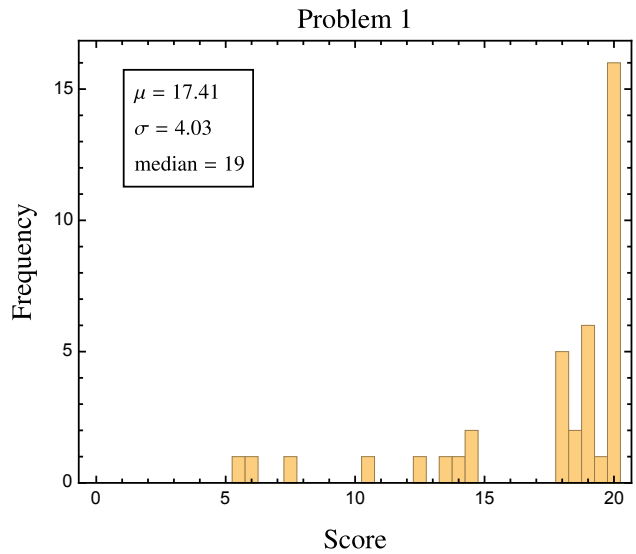


Figure 3: The distribution of scores on Problem 1

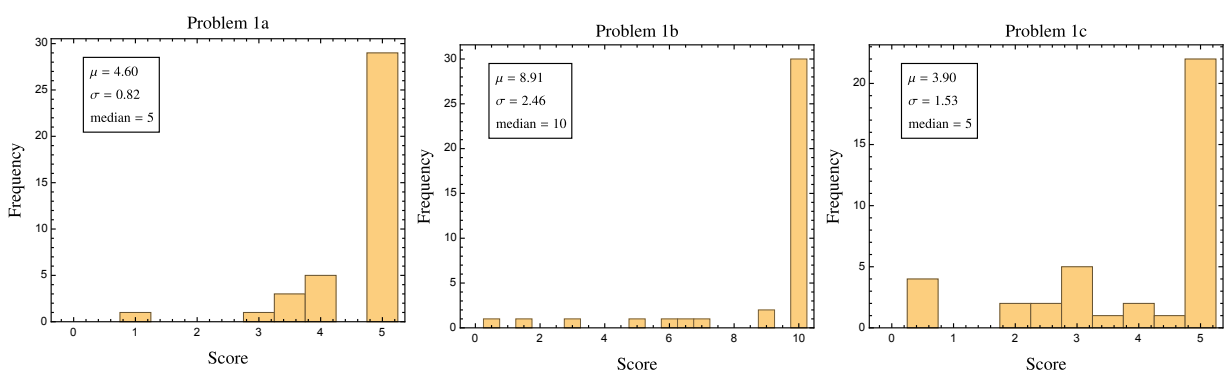


Figure 4: Breakdown of distributions by parts of Problem 1

0.2 Summary Histogram: Problem 2

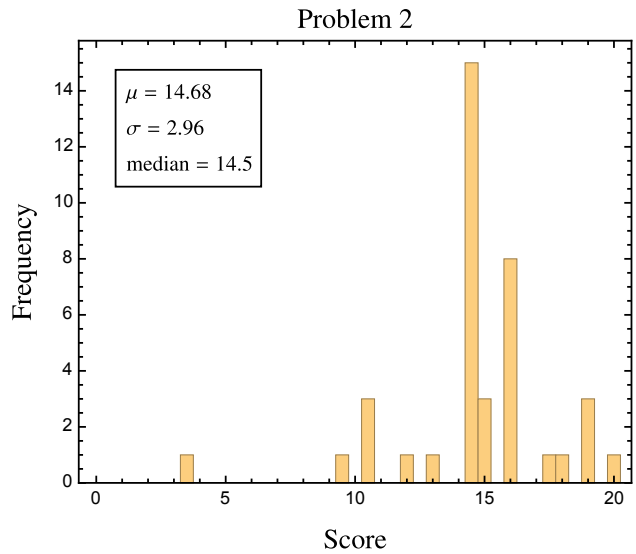


Figure 5: The distribution of scores on Problem 2

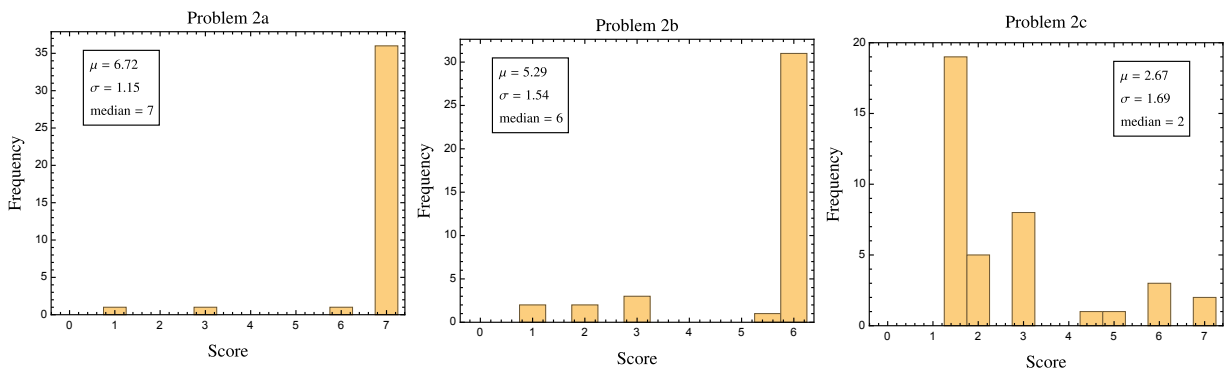


Figure 6: Breakdown of distributions by parts of Problem 2

0.3 Summary Histogram: Problem 3

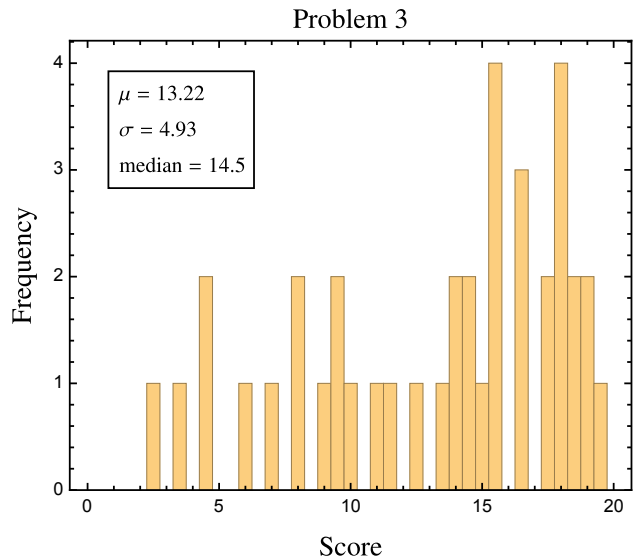


Figure 7: The distribution of scores on Problem 3

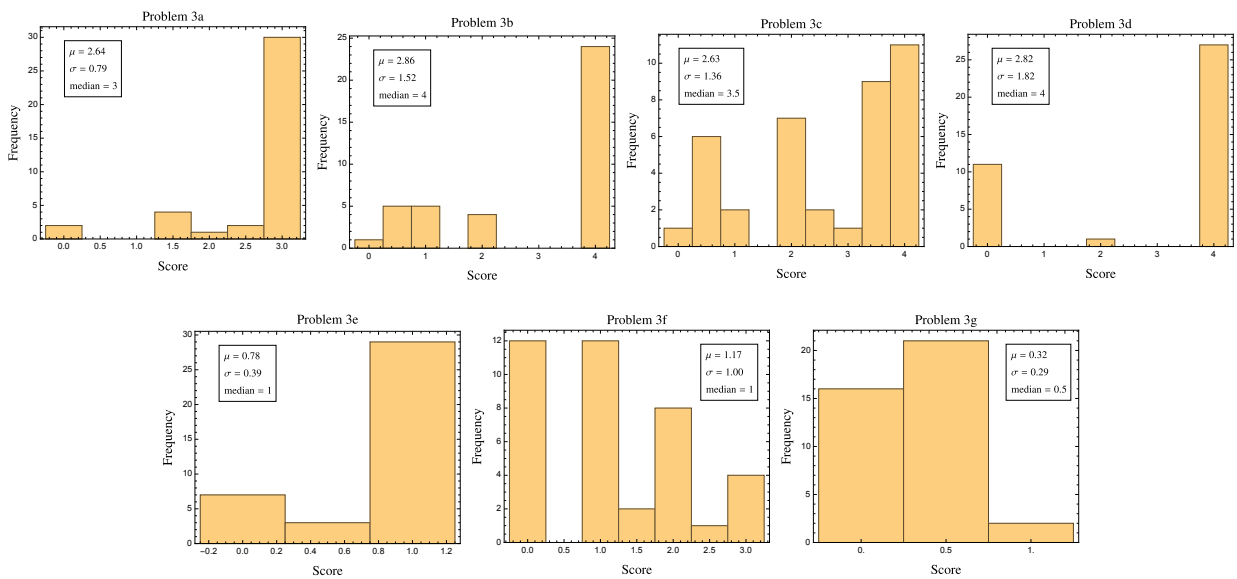


Figure 8: Breakdown of distributions by parts of Problem 3

1 Problem 1

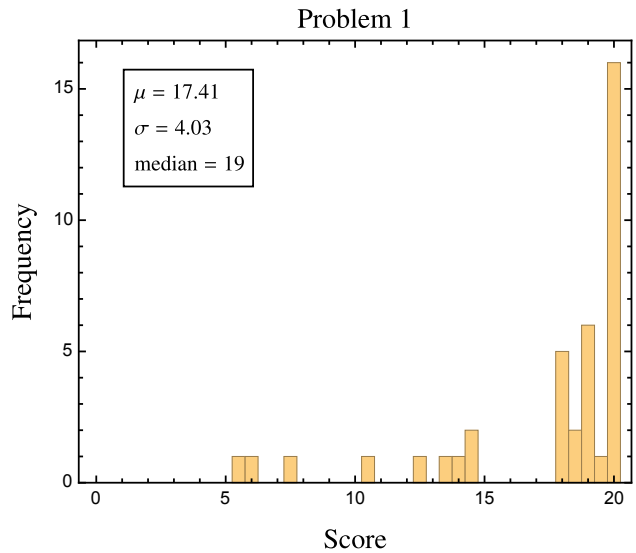


Figure 9: The distribution of scores on Problem 1

1.1 Problem 1(a) *(5 Points)*

5 points	Completely correct
5 points	Sign error in the cross product
4 points	Found the correct magnitude of the curl but didn't specify which the component. Correctly concluded the force was not conservative
4 points	Found the right component was nonzero, but said it was equal to -1 instead of $-k$.
4 points	Found the curl correctly but assumed everything was confined to 2D space so ignored the k-component.
3.5 points	Knew how to take the curl it seems, but accidentally plugged in F_x for F_y and vice versa.
3.5 points	Knew to take a cross product, and seemed to do it correctly, but got 0 for $-d/dy(ky)$ somehow
3 points	Thought the curl was parallel to $\hat{x}$, but found it wasn't zero, concluded force was not conservative correctly.
1 point	Didn't demonstrate knowledge of how to take the curl, found it was zero, consistently but incorrectly assumed force was conservative.
0 points	No answer

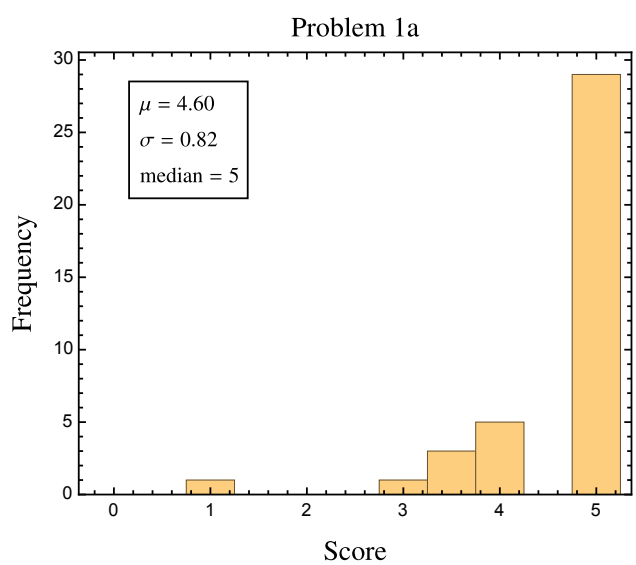


Figure 10: The distribution of scores on Problem 1a

1.2 Problem 1(b) *(10 Points)*

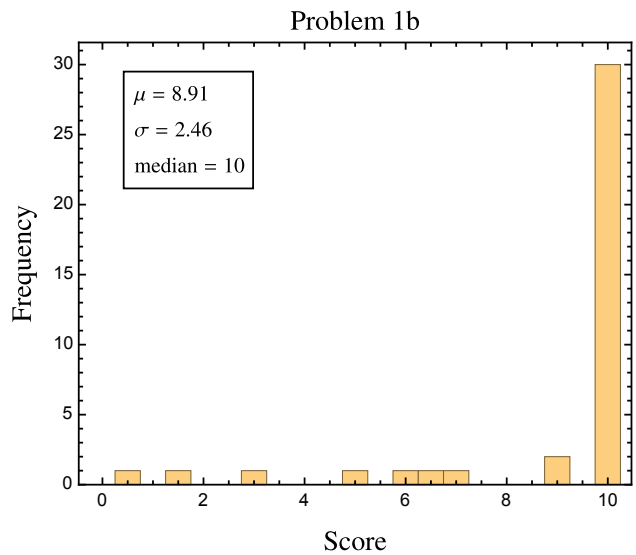


Figure 11: The distribution of scores on Problem 1b

- **10 pts:** Important aspects:
  - (1 pt) Knew to use $W = \int \vec{F} \cdot d\vec{r}$
  - (3 pts) Took the dot product correctly $\int \vec{F} \cdot d\vec{r} = kydx + l\frac{dy}{dx}dy$
  - (1 pt) Substituted $\frac{dy}{dx} = 1$
  - (2 pt) Used $y = x$ to convert the variables in the integrand into a form that they can be integrated (only y's in dy integrals and/or only x's in dx integrals)
  - (1 pt) Performed the integral correctly
  - (1 pt) Used the correct bounds of integration
  - (1 pt) Plugged everything in correctly to get the right final answer

1.3 Problem 1(c) *(5 Points)*

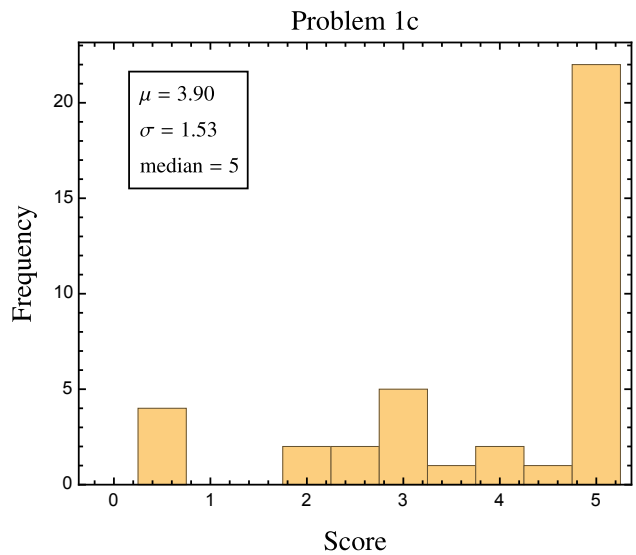


Figure 12: The distribution of scores on Problem 1c

- **5 pts:** Important aspects:
  - (1 pts) Took the dot product correctly $\int \vec{F} \cdot d\vec{r} = kydx + l\frac{dy}{dx}dy$
  - (1 pt) Substituted $\frac{dy}{dx} = 2x$
  - (2 pt) Used $y = x^2$ to convert the variables in the integrand into a form that they can be integrated (only y's in dy integrals and/or only x's in dx integrals)
  - (.5 pt) Performed the integral correctly and used the correct bounds of integration
  - (.5 pt) Plugged everything in correctly to get the right final answer

2 Problem 2

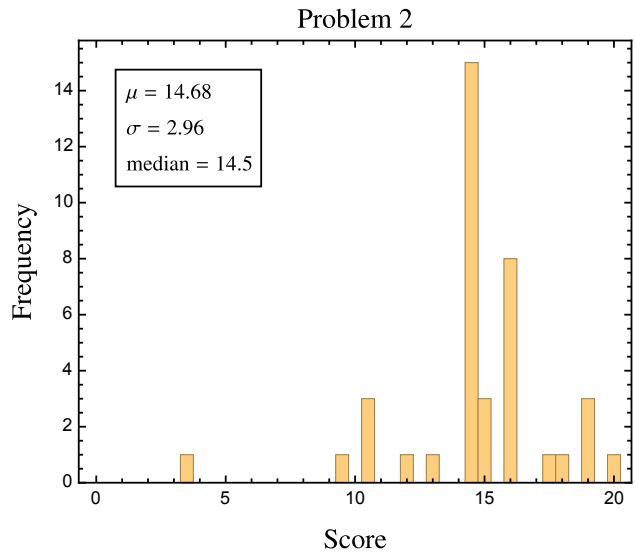


Figure 13: The distribution of scores on Problem 2

2.1 Problem 2(a) *(7 Points)*

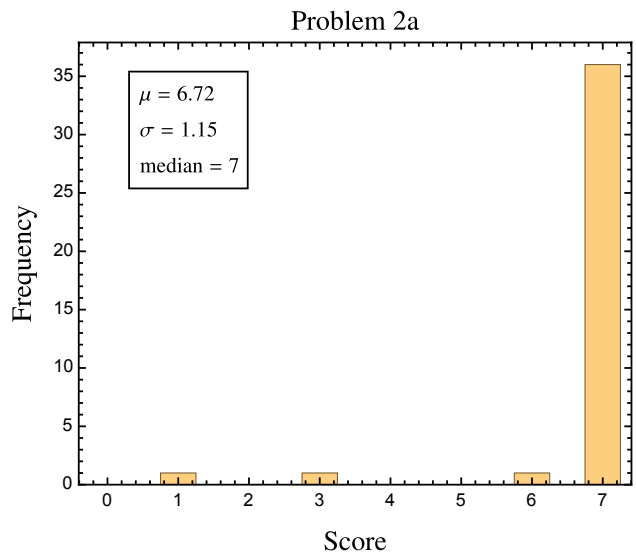


Figure 14: The distribution of scores on Problem 2a

- **7 pts:** Completely Correct
- **6 pts:** Used $\vec{F} = m\vec{a}$ correctly to get the right equation of motion. Guess the right form of a solution $x(t) = e^{rt}$, plugged it back in. From there, made multiple mistakes and arrived at the wrong answer, but it overall had most of the right ingredients.
- **4.5 pts:** Used $\vec{F} = m\vec{a}$ correctly to get the right equation of motion. Guess the right form of a solution $x(t) = e^{rt}$, but never plugged that solution back into the differential equation to find r. Knew two solutions were required but couldn't find them.
- **3 pts:** Used $\vec{F} = m\vec{a}$ correctly to get the right equation of motion. Tried to solve the differential equation incorrectly from there.
- **3 pts:** Used $\vec{F} = m\vec{a}$ correctly to get the right equation of motion, but stopped there.
- **3.5 pts:** Forgot the $m\vec{a}$ part of Newton's second law, and instead wrote $\vec{F}_{\text{total}} = 0$. As a result, didn't get the $\ddot{x}$ part of the differential equation. Still guessed $x = e^{rt}$ and tried to get a characteristic equation, looked for two solutions and in the end had to engineer a second incorrectly. Got an answer in the right form but with the wrong ω_1.
- **2 pts:** Added a driving force to the problem that wasn't there in the setup, then proceeded to find the homogeneous solution. Wrote down the characteristic equation from memory (no derivation), then found the root of that equation. Did not plug that root in correctly to $x(t)$, and $x(t)$ didn't have any t-dependence in the end.

- **1.5 pts:** Forgot the $m\vec{a}$ part of Newton's second law, and instead wrote $\vec{F}_{\text{total}} = 0$. As a result, didn't get the $\ddot{x}$ part of the differential equation. Solved the first order differential equation.

- Partial Credit Scheme to assign points above
  - **3 pts:** Using Newton's 2nd law to obtain the EOM
  - **1 pts:** Realizing that the EOM required an exponential form to solve
  - **1 pts:** Plugging the exponential form back in to obtain the characteristic equation
  - **.5 pts:** Finding the roots of the characteristic equation
  - **1.5 pts:** Adding the two linearly independent solutions together with arbitrary coefficients to obtain the full general answer

2.2 Problem 2(b) *(6 Points)*

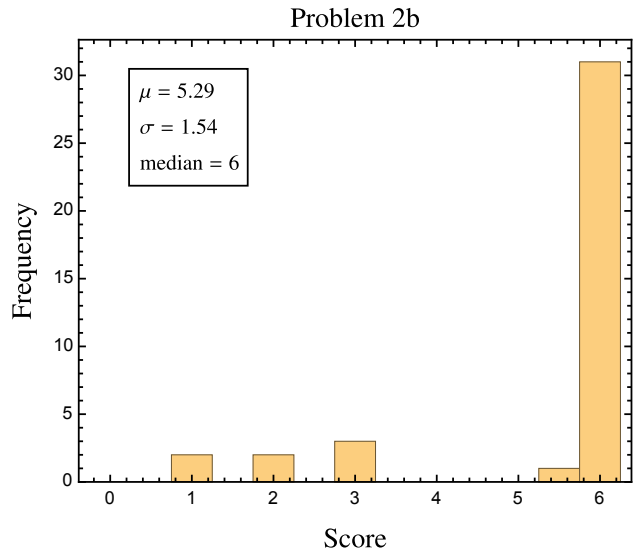


Figure 15: The distribution of scores on Problem 2b

- **6 pts:** Completely correct
- **5.5 pts:** Plugged in k incorrectly and concluded that the system was critically damped.
- **5.5 pts:** Plugged in β incorrectly and concluded that the system was critically damped.
- **5.5 pts:** Made an algebra error reducing ω_0 but correctly identified the conditions for oscillation. Final statement consistent given the mistake.
- **5 pts:** Made a mistake in reducing algebra for ω_0, but ended up getting $\beta < \omega_0$ without good cause, and deduced that the system was therefore oscillating.
- **4 pts:** Correctly compared β and ω_0, from memory realized this means the system was underdamped and oscillating. Didn't connect this to the argument of the exponent though (mostly due to mistakes in the previous part)
- **3.5 pts:** Knew something inside a square root had to be negative to give oscillation. Tried to reverse-engineer the ω_1 given in part (c), but didn't realize this was already after the imaginary piece had been pulled out. Concluded the system wasn't oscillating.
- **3 pts:** Knew something inside a square root had to be negative to give oscillation. Didn't really explain why because never found the exponential solution above. Tried to compare k with ω_0^2.
- **3 pts:** Correctly compared β and ω_0, no mistakes in plugging things in. Did not know whether this meant the system was oscillating or not, or incorrectly claimed this means the system is not oscillating.

- **3 pts:** Correctly compared β and ω_0, no mistakes in plugging things in. Incorrectly claimed this meant $x(t)$ had imaginary components.
- **2.5 pts:** Knew that the crux of the question was complex exponents, but incorrectly stated that complex exponents don't give oscillation. Never compared β and ω_0.
- **2.5 pts:** Concluded that the system was not oscillating because v doesn't change sign. This was correct given previous errors. These errors, however, simplified the problem significantly.
- **2 pts:** Concluded that the system was not oscillating because the damping force wasn't zero
- **2 pt:** Said the system was oscillating without an explanation related to the physics of the problem
- **1 pt:** Said the system was not oscillating without an explanation related to the physics of the problem

2.3 Problem 2(c) *(7 Points)*

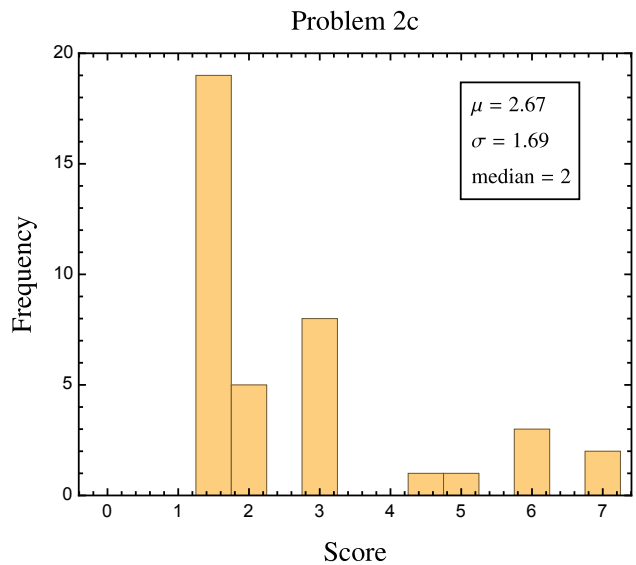


Figure 16: The distribution of scores on Problem 2c

Partial Credit Scheme for answers largely on the right track

- **3 pts:** Use energy conservation $\Delta E = \Delta T + \Delta U = W_{\text{drag}}$
- **1 pts:** Realize $\Delta T = 0$
- **1 pts:** Realize $\Delta U = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_0^2$
- **1 pts:** Find x_1 in terms of x_0 in order to find ΔU
- **1 pts:** Correctly simplify (reduce factors of $e^{2\pi i}$ for example)

Partial Credit Scheme for answers going in very different directions

- **1.5 pts** : Wrote down something related to W that was physically relevant, even if you didn't get the right answer
- **1.5 pts** : Plugged in initial conditions correctly to simplify your work

3 Problem 3 *(20 Points)*

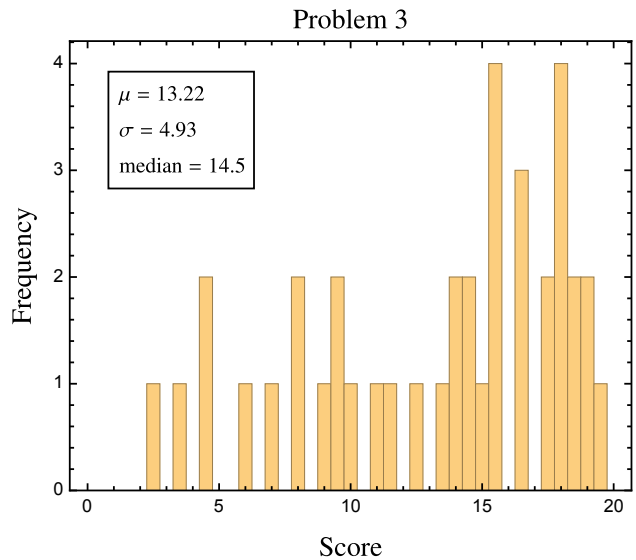


Figure 17: The distribution of scores on Problem 3

3.1 Problem 3(a): Find β, ω_0 *(3 Points)*

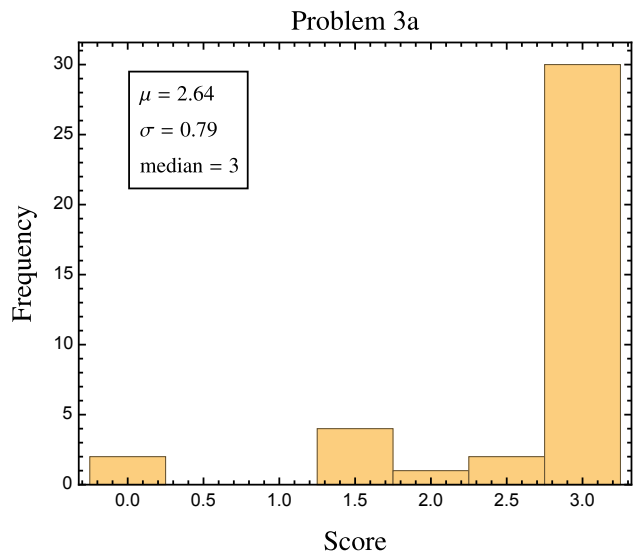


Figure 18: The distribution of scores on Problem 3a

- **1.5 pts:** Found β correctly
- **1.5 pts:** Found ω_0 correctly
- Partial credit available in either category if they got close to the right answer but just dropped a factor.

Figure 19: The distribution of scores on Problem 3a

3.2 Problem 3(b): Find a_0 *(4 Points)*

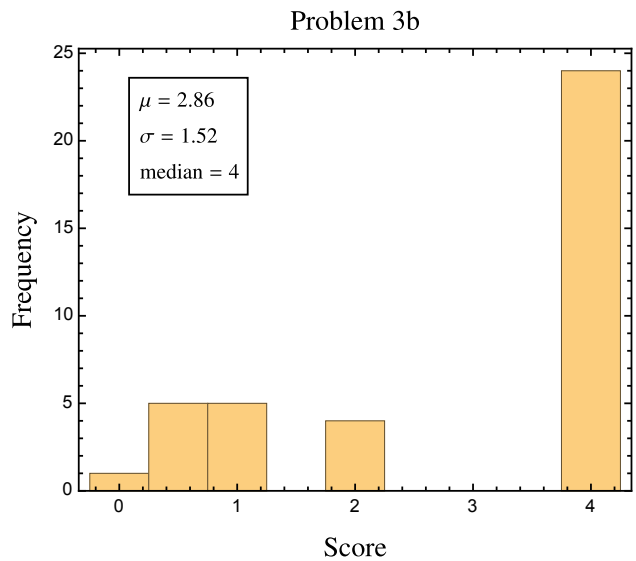


Figure 20: The distribution of scores on Problem 3b

- **4 pts:** Completely correct. If the units were wrong, they showed their work so I knew how they got to the answer.
- **3.5 pts:** Dropped the $1/\tau$ halfway through
- **3 pts:** Got the right numerical answer with the wrong or missing units, and didn't show any work to explain how they got there.
- **3 pts:** Found the integral correctly, divided by τ, but forgot to multiply by 2 at the end.
- **3 pts:** Thought $\tau = 3$ s or $\tau = 4$ s, otherwise did everything correctly.
- **2 pts:** Used the maximum of $f(t)$ in the integrand rather than $f(t)$
- **2 pts:** Didn't read $f(t)$ correctly off the plot, otherwise everything was correct.
- **1 pts:** Didn't know how to put $f(t)$ in from the graph, instead used the Fourier Series equation again
- **0.5 pts:** Wrote something relevant down

Figure 21: The distribution of scores on Problem 3b

3.3 Problem 3(c): Find a_n *(4 Points)*

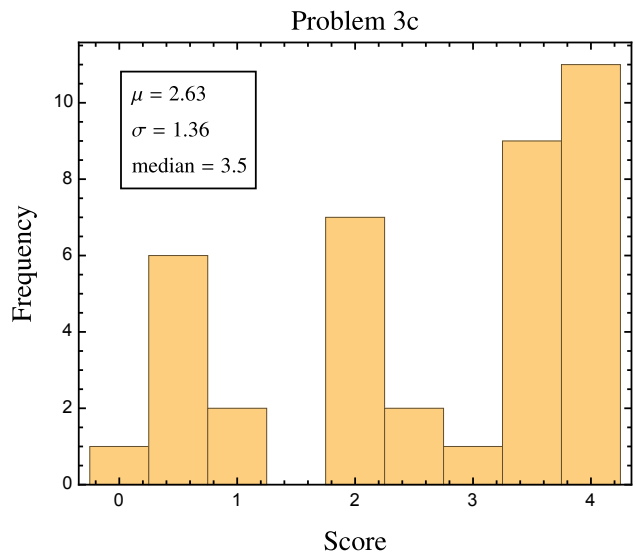


Figure 22: The distribution of scores on Problem 3c

Partial credit scheme:

- **2 pts:** Knew how to plug in $f(t)$ by reading it off the graph
- **2 pts:** Integrated and reduced algebra correctly and got the right answer. Partial credit available

Figure 23: The distribution of scores on Problem 3c

3.4 Problem 3(d): Find b_n *(4 Points)*

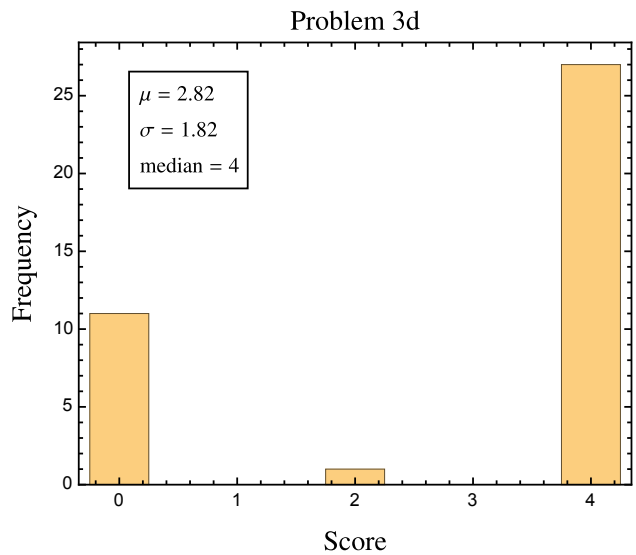


Figure 24: The distribution of scores on Problem 3d

- **4 pts:** Correct answer $b_n = 0$ arrived at by any means
- **0 pts:** Blank or wrong

Figure 25: The distribution of scores on Problem 3d

3.5 Problem 3(e): Fill in the blank for $f(t)$ *(1 Points)*

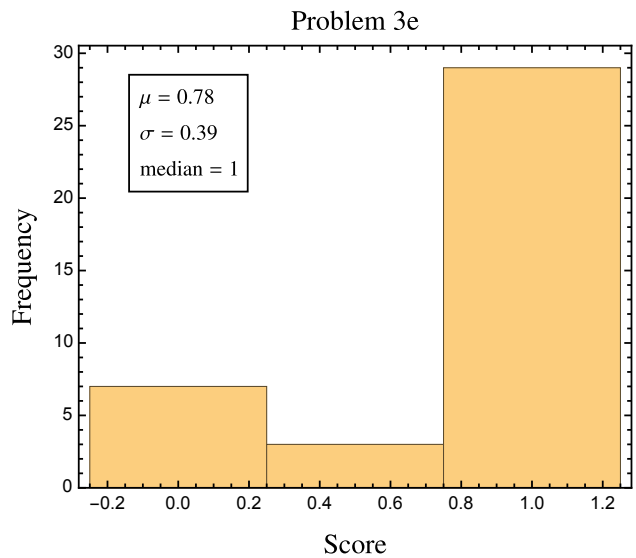


Figure 26: The distribution of scores on Problem 3e

Partial credit scheme:

- **1 pts:** Completely correct (or consistent with previous answers)
- **0.5 pts:** Forgot the $\cos(n\omega t)$ inside the sum
- half point partial credit taken off/added where appropriate

Figure 27: The distribution of scores on Problem 3e

3.6 Problem 3(f): Find $x_n(t)$ *(3 Points)*

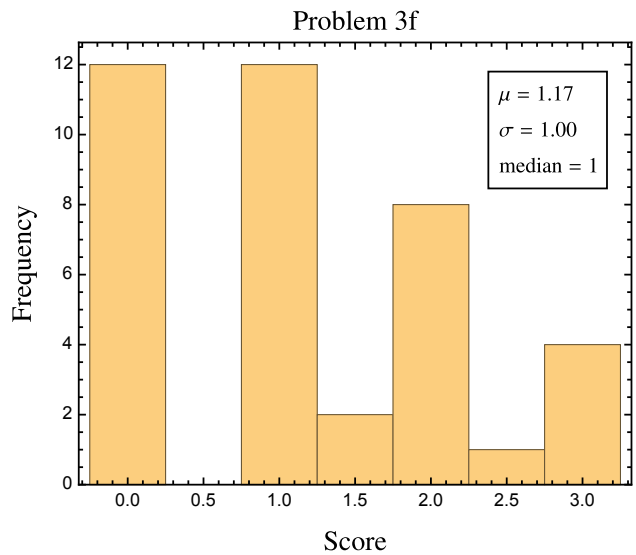


Figure 28: The distribution of scores on Problem 3f

Partial credit scheme:

- **1 pts:** Getting the right form of $x_n(t)$
- **1 pts:** Realizing the A, δ on the formula sheet should be used
- **1 pts:** Substituting $\omega \to n\omega$ appropriately to obtain A_n, δ_n. Realizing $f_0 = a_n$.

3.7 Problem 3(g): Find $x(t)$ *(1 Points)*

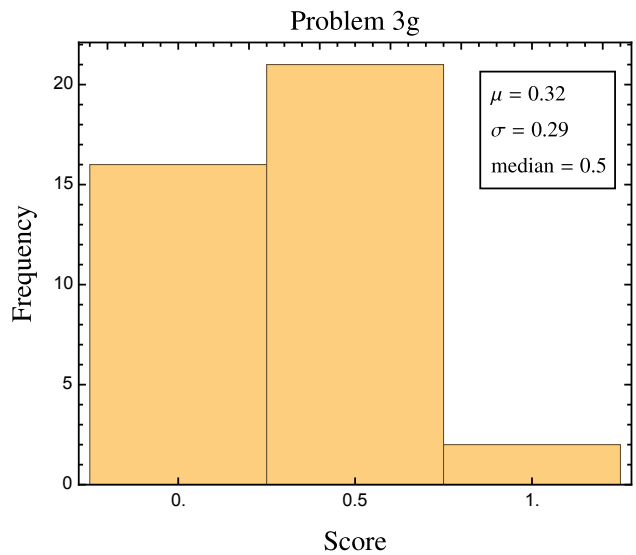


Figure 29: The distribution of scores on Problem 3g

- **1 pts:** Correct or consistent answer
- **0.5 pts:** Forgot to add in the homogeneous equation
- **0.5 pts:** Forgot to add in the $n = 0$ case separately
- **0.5 pts:** Forgot to add in the $n = 0$ case separately and forgot the homogeneous equation
- **0 pts:** Wrong or blank